

SPECIALIZED PROJECT (CO4029)

LEVERAGING MULTIMEDIA DATA FOR NEXT BEST ACTION RECOMMENDATION SYSTEM

ADVISOR: PhD. LE THANH VAN

STUDENT 1 : DANG MINH KHANG (2252287)

STUDENT 2: PHAM TRAN DANG KHOA (2252360)

STUDENT 3: LE HA NGUYEN KHANH (2252327)

Table of Contents

1. Project Objectives & Scope
2. Problem Statement & Motivation
3. Foundational Knowledge
4. Exploratory data analysis
5. Model evaluation
6. System Design & Architecture
7. Achievements & Future Work Plans

1

PROJECT OBJECTIVE & SCOPE

PROJECT OBJECTIVE

Foundational Knowledge

1. Multimodal
2. Contrastive Learning
3. Encoder Architectures



System Architecture

1. Modular Decision Layers
2. Vector Storage
3. NBA Decision Engine



Problem Solving

1. Text Encoders
2. Vision language Model
3. Fusion Strategies



PROJECT SCOPE

IN SCOPE

1. Theory & literature
2. Amazon Reviews 2023 analysis
3. Encoder evaluation
4. System design
5. Proof of concept

LIMITATIONS

1. English only
2. Frozen encoders
3. Dev environment
4. Books domain

FUTURE WORK

1. Larger datasets
2. Full system build
3. Web demo
4. Deployment & evaluation

2

PROBLEM STATEMENTS & MOTIVATION

THE PROBLEM: INFORMATION OVERLOAD

Global Ebook Market



\$16.7B USD^(*)

Millions of titles available

The Question

How do users find the **RIGHT** book **efficiently**?

Traditional Approaches Fall Short

- **Collaborative filtering:** cold start problem
- **Content based:** single modality
- **Metadata only:** misses context

^(*) 6Wresearch. "How Big Is the eBook Market?" url : [6wresearch.com/market-takeaways-view/how-big-is-the-ebook-market](https://www.6wresearch.com/market-takeaways-view/how-big-is-the-ebook-market)

NEXT BEST ACTION SYSTEM CONCEPT

What is NBA?

Next Best Action goes beyond simple recommendations:

- Not just *"What book might you like?"*
- But ***"What specific action should we suggest right now?"***

Key Insight: Different user contexts require different action strategies

Traditional	Return Top-K items
NBA	Choose the most appropriate response strategy

Types	Actions
DIRECT_RECOMMENDATION	High-confidence single item
SERIES_CONTINUATION	Complete the trilogy
CURATED_COMPARISON	Top 3 with pros/cons
VISUAL_SIMILARITY	Books with similar cover aesthetics
DIVERSE_EXPLORATION	Cross-genre suggestions

WHY MULTIMODAL FOR BOOKS?

Text Modality

1. **Detailed** reviews (average **142 tokens**)
2. **Rich** metadata (author, genre, theme)
3. **Captures**: plot, writing style, tone

Example: "Mystery novel, strong female lead, minimalist style"

Image Modality

1. **High quality** book covers (**99.1%**)
2. **Genre** conventions (colors/typography)
3. **Captures**: aesthetics, visual patterns

Example: "Find books with covers similar to dark fantasy"

Multimodal Advantage

INSIDE (content) + what's **OUTSIDE**
(aesthetics)

More comprehensive understanding
of user preferences

03

FOUNDATIONAL KNOWLEDGE

FOUNDATIONAL KNOWLEDGE: CONTRASTIVE LEARNING

Core Principle:

Learn representations by **maximizing similarity between positive pairs** and **minimizing similarity between negative pairs** in a shared embedding space.

InfoNCE Loss Function:

$$L_{InfoNCE} = -\log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\exp(\text{sim}(z_i, z_i^+)/\tau) + \sum_{j=1}^{N-1} \exp(\text{sim}(z_i, z_j^-)/\tau)}$$

Where:

- z_i : query embedding
- z_i^+ : positive example
- z_j^- : negative examples
- τ : temperature parameter
- $\text{sim}(\cdot, \cdot)$: Similarity function

ENCODER ARCHITECTURES: BLaIR

BLaIR (Bridging Language and Items for Retrieval)

Strength: Deep semantic understanding of textual content

Why BLaIR?

1. **Pre-trained on Amazon Reviews** → domain-specific
2. **Understands** e-commerce language patterns
3. **Captures:** product descriptions, user reviews, metadata

Training Approach

1. **Contrastive learning on user item interactions**
2. **Query product** matching task
3. **Optimized** for **semantic** similarity



Architecture

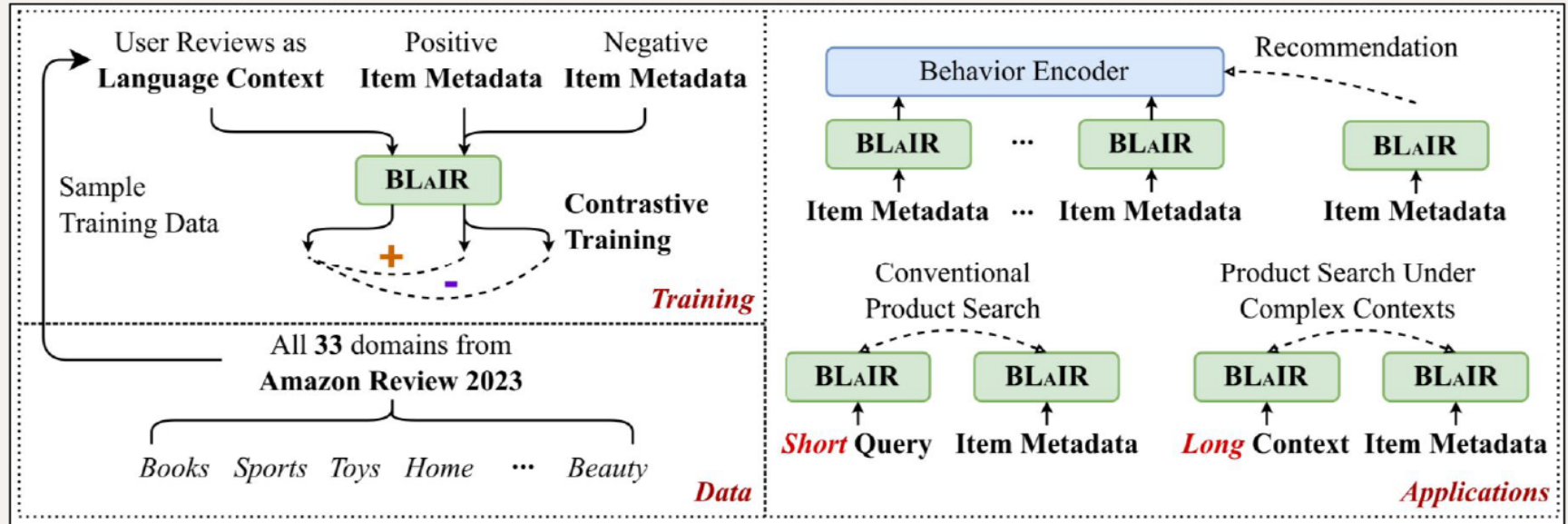
RoBERTa-Large
(24 layers,
355M parameters)



Dimension

1024-dimension
encoding

ENCODER ARCHITECTURES: BL_AIR



Yupeng Hou, Jiacheng Li, Zhankui He, An Yan, Xiushi Chen, and Julian McAuley. Bridging language and items for retrieval and recommendation, 2024. URL <https://arxiv.org/abs/2403.03952>.

ENCODER ARCHITECTURES: CLIP

CLIP (Contrastive Language-Image Pre-training)

Strength: Vision-language understanding, genre recognition from covers

Dual-Encoder Design

1. **Image Encoder:** Vision Transformer
2. **Text Encoder:** Modified Transformer
3. **Shared** embedding space (512-d)

Pre-training Objective

1. **400M** image-text pairs from internet
2. **Contrastive learning** across modalities
3. **Strong Zero-shot** abilities



Architecture

Vision Transformer (ViT-B/32)

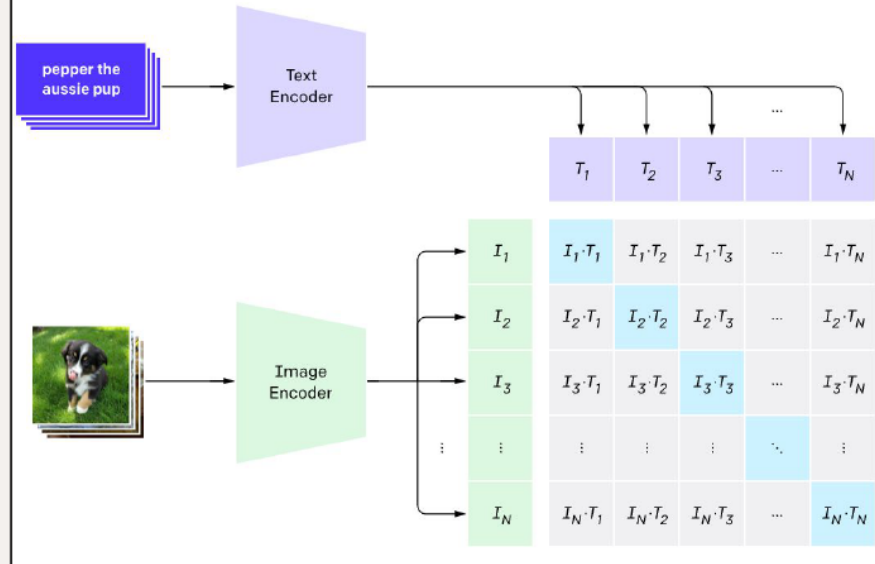


Dimension

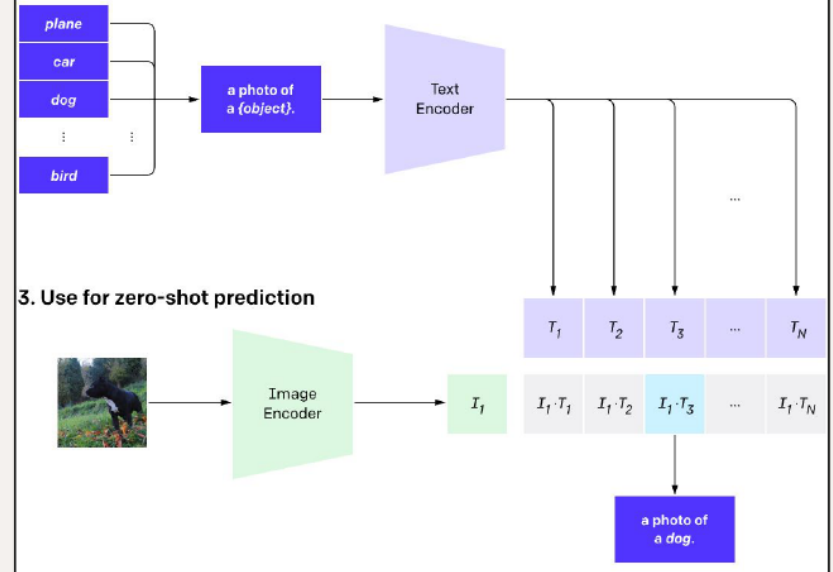
512-dimension encoding

ENCODER ARCHITECTURES: CLIP

1. Contrastive pre-training



2. Create dataset classifier from label text



3. Use for zero-shot prediction

OpenAI, "CLIP: Connecting text and images," January 5, 2021, <https://openai.com/index/clip/>

FUSION STRATEGIES COMPARISON

Early

Mid

About

- ❑ **Concatenate** features **before** encoding
- ❑ **Single joint** representation

- ❑ **Intermediate** layer fusion
- ❑ **Cross attention** mechanisms

Drawbacks

Loses **modality-specific** info

Complex training

FUSION STRATEGIES: LATE FUSION



Separate encoders

For each modality



Combine

At decision layer

$$score_{final} = \beta \times score_{BLaIR} + (1 - \beta) \times score_{CLIP}$$

Optimal: $\beta \approx 0.8$ (Text-dominant by design)

Advantages

1. Modular architecture
2. Handles missing modalities
3. Tunable fusion weights
4. Preserves modality strengths

4

EXPLORATORY DATA ANALYSIS

DATASET: AMAZON REVIEWS 2023

McAuley-Lab/Amazon-Reviews-2023

1. Statistics

- **Total Reviews:** ~571 million across all categories
- **Books Category:** ~29 million reviews
- **Time Period:** May 1996 – September 2023

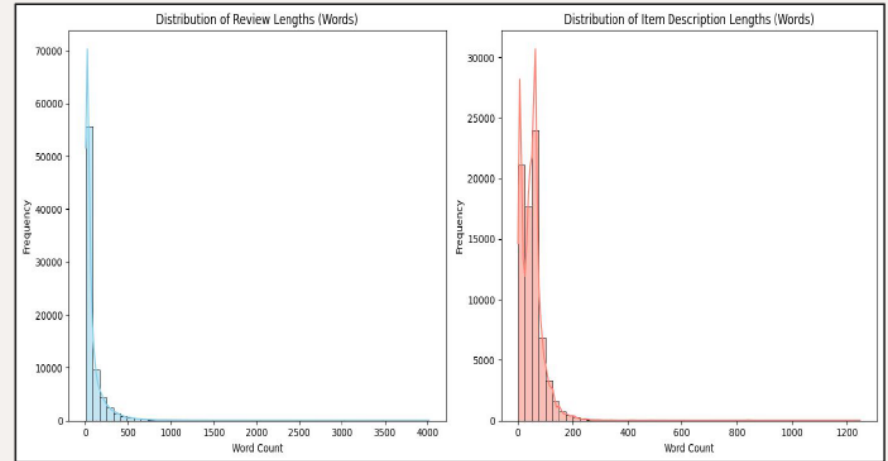
2. Dataset Descriptions

METRIC	RAW DATASET	WORKING DATASET
Review Records	29,475,453	76,347 (merged)
Item Metadata	4,448,181	38,654 books
Users	—	34,048 users

TEXT LENGTH DISTRIBUTIONS

METRIC	REVIEW TEXT	ITEM DESCRIPTION
Mean (words)	86.2	51.9
Median (words)	37.0	50.0
Maximum (words)	4,013	1,250
90th Percentile	204	~100

Text length histogram



DESIGN DECISION: TRUNCATION POLICY

- **BLaIR Input Limit: 512 tokens** (~350-400 words)
- **Truncate from END** to preserve key information
- Only top **10%** of reviews **require truncation**

Multimodal Asymmetry in Image Availability



98.2% Reviews

Have No Images



99.1% Items

Have Cover Images

INSIGHTS

- The 99% vs 2% image asymmetry resolves architectural choice: the system is built on **catalog data**, not user uploads.
- Text supports **100% of queries**, while cover images provide **reliable visual augmentation** for ~99% of items.
- Multimodal fusion is evaluated where both signals exist, yielding a **+12.7% gain**, proving vision is **complementary, not redundant**.
- User behavior confirms the design: reviews are text-centric; images are rare and situational.

CONCLUSION

Two empirical findings eliminate design ambiguity:

- **Image Asymmetry (99% vs 2%)** → Multimodal retrieval must be built on **item-side data** (text + covers), not user uploads
- **Text Distribution ($90\% \leq 512$ tokens)** → Truncation is **natural, efficient, and semantically safe**



Universal

Works for 100% of queries



Robust

Near-universal visual signal from covers



Scalable

No dependency on sparse user behavior

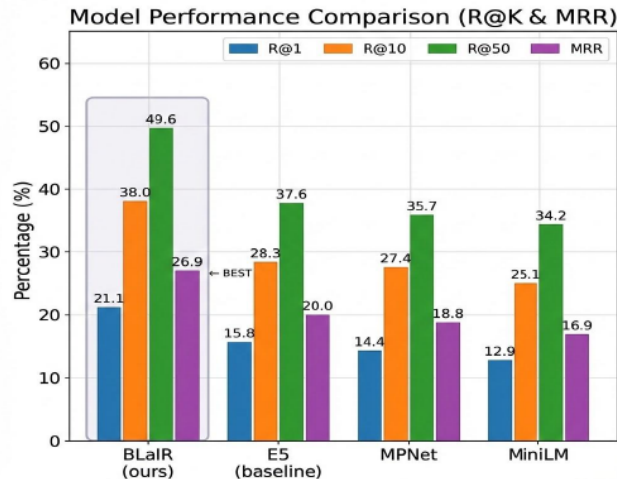
5

MODEL EVALUATION

BASELINES & Why BLaIR Outperforms General Models ?

Model	R@1	R@10	R@50	MRR
BLaIR (RoBERTa-large)	0.2106	0.3802	0.4960	0.2688
E5 (BERT-base)	0.1584	0.2829	0.3759	0.1997
MPNet (base)	0.1443	0.2738	0.3567	0.1877
MiniLM (distilled)	0.1289	0.2513	0.3422	0.1692

**WHY
BLaIR
WINS**



1. DOMAIN-SPECIFIC TRAINING

- **Trained** on Amazon product data
- **Learns** e-commerce vocabulary & patterns
- **33% improvement** over E5

2. ARCHITECTURAL CAPACITY

- **24 layers** (vs 12 for baselines)
- **1024-dim** embeddings (vs 768)
- **Handles** attribute-rich product text

3. CONTRASTIVE LEARNING

- **Optimized** for **retrieval** (not just MLM)
- **Implicit** visual grounding from **multimodal** data
- **Better embedding** geometry for similarity

VISUAL SIGNAL: CLIP STANDALONE

Setting	R@1	R@10	R@50	MRR
CLIP Text-only ($q_{\text{text}} \rightarrow v_{\text{text}}$)	0.2200	0.4212	0.6115	0.2901
CLIP Image-only ($q_{\text{img}} \rightarrow v_{\text{img}}$)	0.3838	0.5663	0.7613	0.4483
CLIP Fused (weighted, $\alpha = 0.45$)	0.4165	0.6615	0.8222	0.4968
CLIP Fused (concat)	0.4649	0.6973	0.8471	0.5439

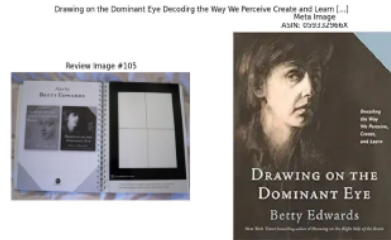


Figure 5.9: Correct CLIP retrieval example 1: Drawing instructional book. The review image shows hand-drawn sketches that match the instructional drawing theme of the retrieved items.

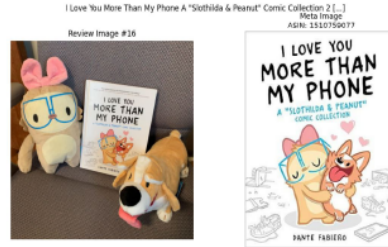


Figure 5.10: Correct CLIP retrieval example 2: Comic collection. The review image displays cartoon-style character illustrations matching the whimsical aesthetic of comic books.

Evaluation Setting

- **Review-item pairs** with both user photos and item covers
- **User image $\rightarrow$ item image** retrieval
- **Hardest visual case:** noisy user photos vs clean catalog covers

Why Concat Fusion Works Best

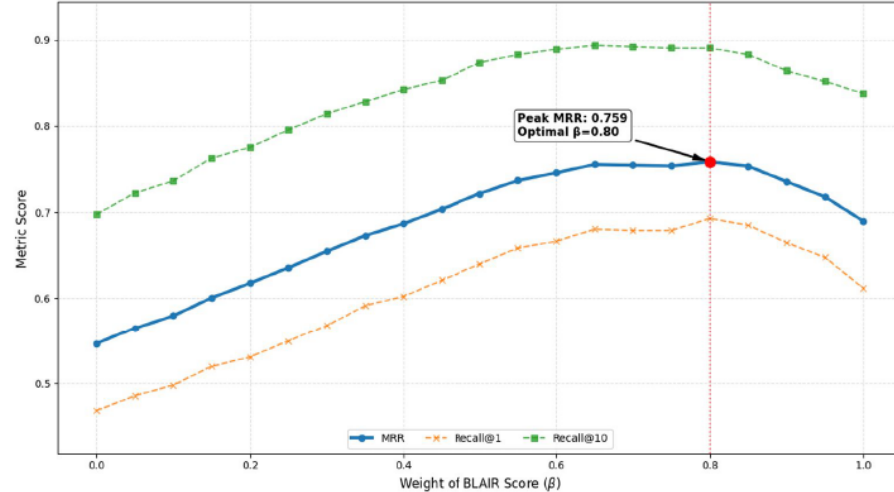
- Preserves **full semantic capacity** of both modalities
- **Avoids premature** information loss from weighting
- **Allows similarity** search to exploit complementary cues:
 - **Text:** titles, keywords, edition hints
 - **Image:** layout, typography, visual identity

Text **disambiguates** meaning, images **refine** identity.

MULTIMODAL FUSION: BLAIR + CLIP

Model	R@1	R@10	R@50	MRR
BLAIR-only	0.615	0.841	0.942	0.691
BLAIR+CLIP (concat, 2048-d)	0.635	0.874	0.952	0.717

Score Fusion Performance (100k Distractors): BLAIR vs CLIP



Observations

- **BLAIR-only:** 61.5% R@1, 0.691 MRR
- **BLAIR + CLIP (concat):** 63.5% R@1, 0.717 MRR
- **BLAIR + CLIP (beta = 0.8):** 69.3% R@1, 0.759 MRR

Higher fusion score

→ **Higher** decision confidence

→ **Different NBA action**

FINAL TAKEAWAYS & SYSTEM DECISIONS

What We Learned

- **Domain text matters:** BLaIR outperforms general encoders
- **Vision helps:** CLIP is robust even on noisy user images
- **Fusion works:** Text + vision consistently improves retrieval
- **Robustness confirmed:** Stable across query types and scale

Evidence-Based Decisions

- **Text Encoder:** BLaIR (**domain-trained**, high-capacity)
- **Visual Encoder:** CLIP (catalog covers, **99% coverage**)
- **Fusion Strategy:** Weighted fusion with **9% better** retrieval
- **Deployment Mode:** Text-first, vision-enhanced retrieval

6

System Design & Architecture

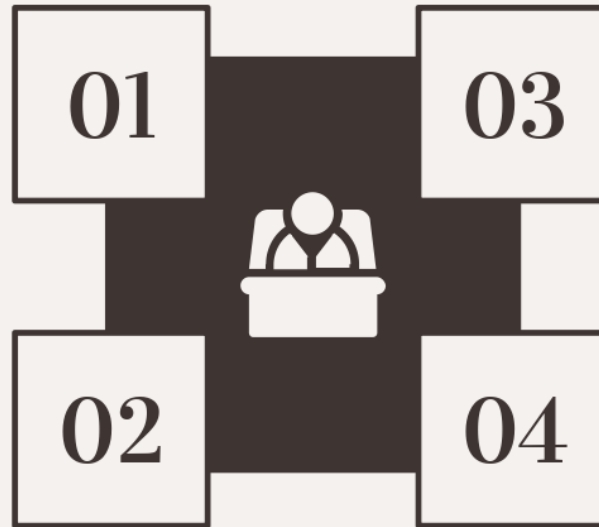
Core System Components

FastAPI Orchestrator

- Input validation
- Routing requests
- Modality detection

Embedding Services

- BLAIR for text semantics
- CLIP for visual semantics
- Independent & replaceable



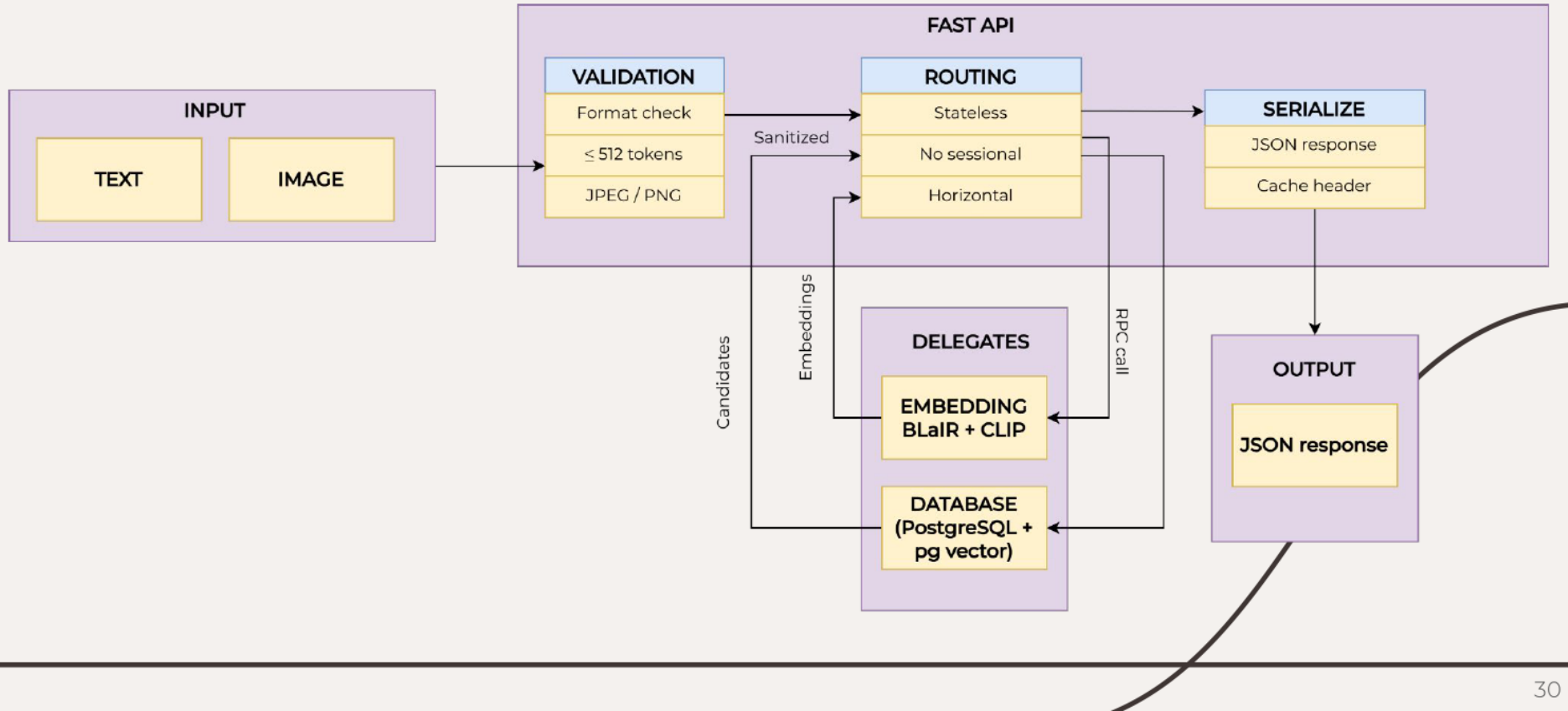
Vector Store (PostgreSQL + pgvector)

- Stores text & image embeddings
- Efficient similarity search

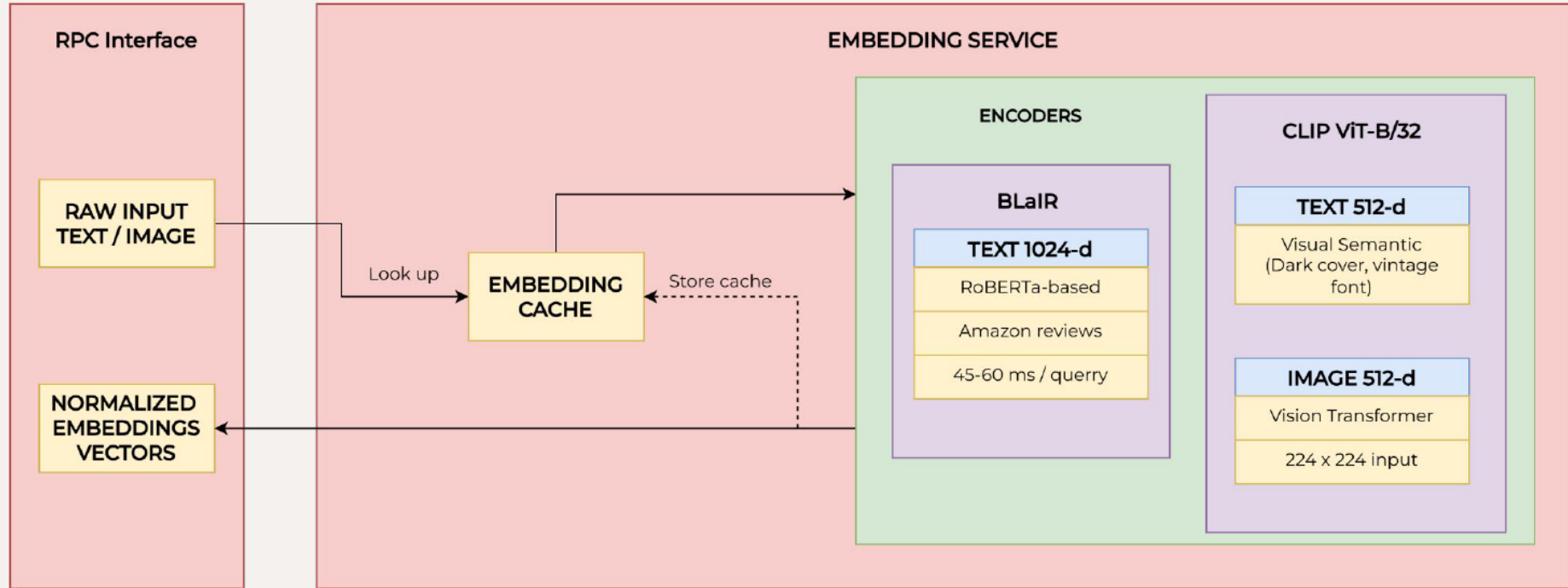
NBA Decision Engine

- **Inputs:** fusion score, score gap, metadata
- **Output:** action type

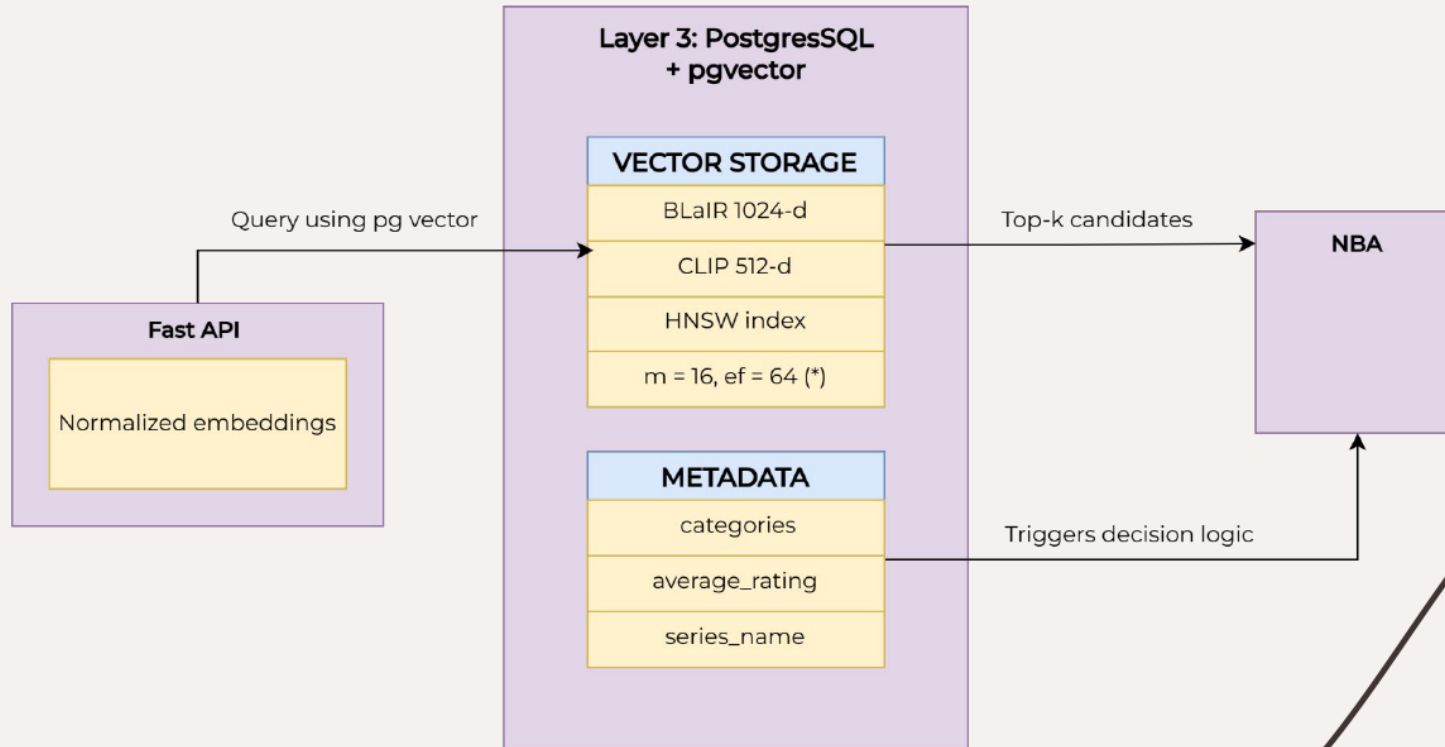
Layer 1: FastAPI Orchestrator



Layer 2: Embedding Service

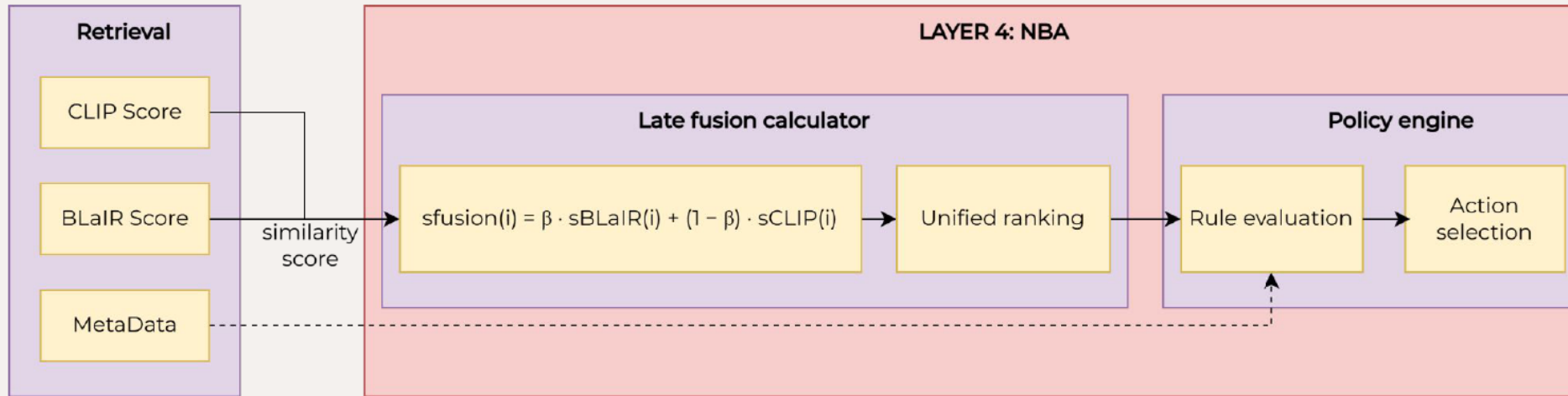


Layer 3: Data layer (PostgreSQL + pg vector)

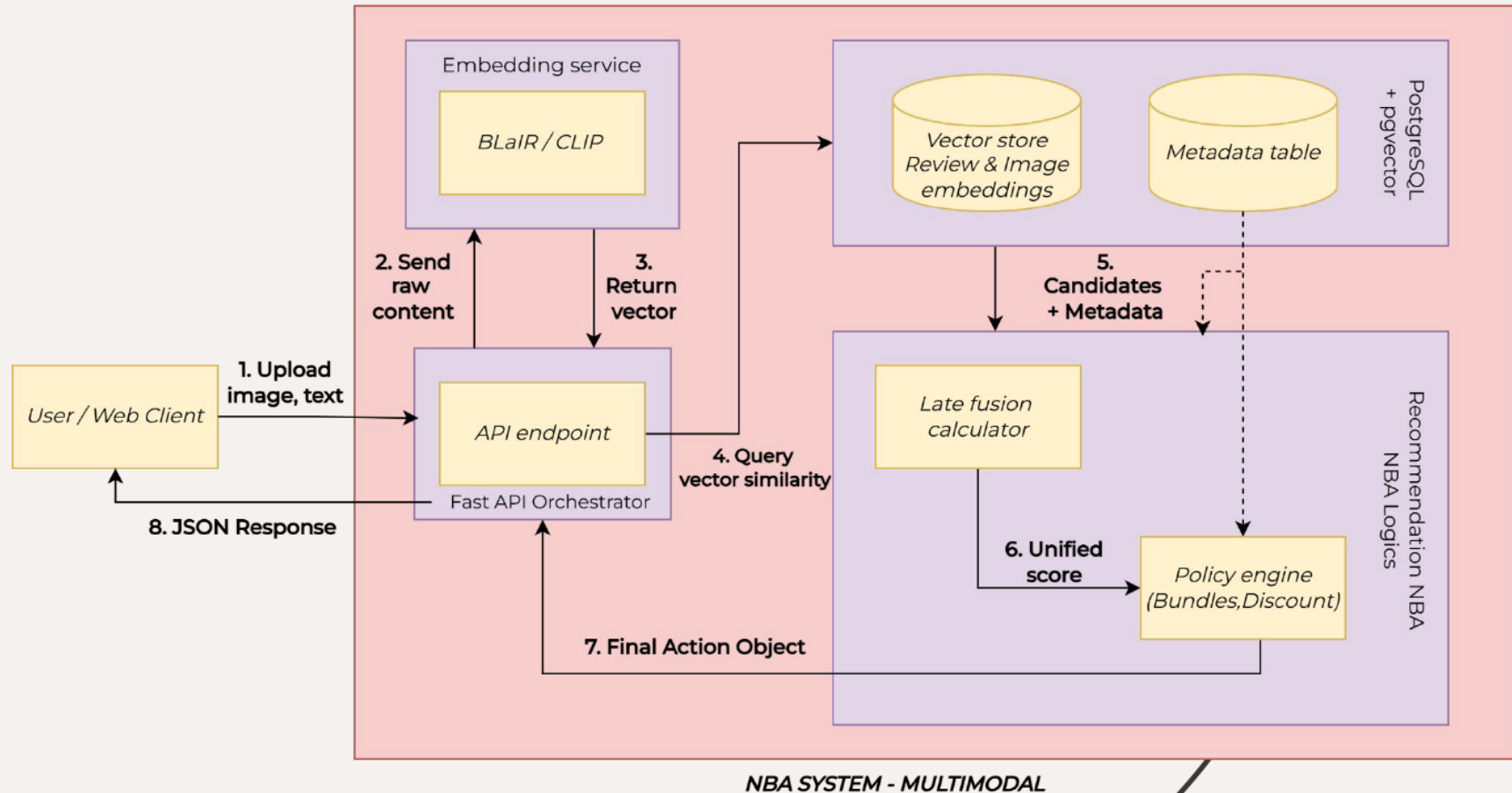


(*) The *m* parameter controls graph connectivity (higher values improve recall but increase index size), while *ef*_construction determines build-time accuracy

Layer 4: NBA Decision Engine



Overall architecture



Design Trade-offs & Limitations

Decisions

- **Frozen encoders** → faster development
- **Rule-based** NBA → interpretability
- **pgvector** → simplicity & reproducibility

Limitations

- No **online learning**
- No **personalization**
- **Single-domain** (books)

7

ACHIEVEMENTS AND FUTURE PLAN

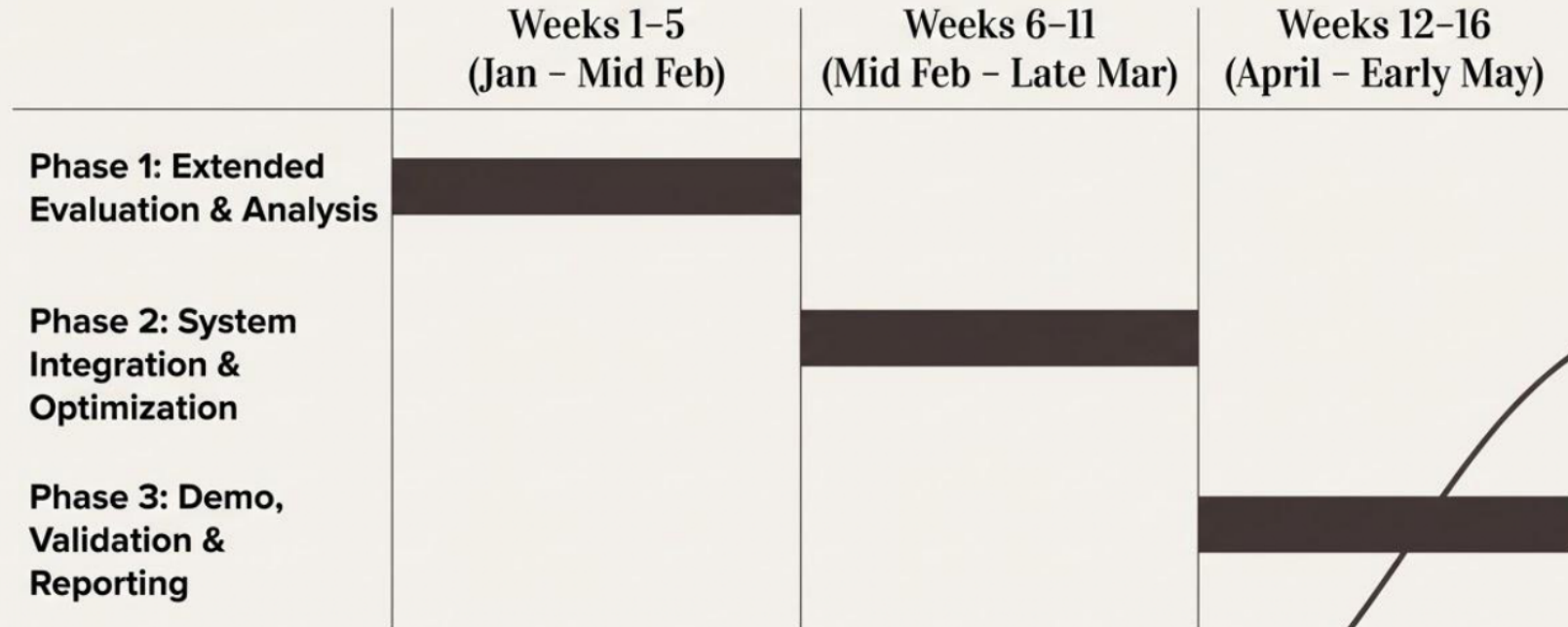
What we achieve

- **Strong theoretical foundations in multimodal recommendation**
- **Conducted EDA and benchmarking on Amazon Reviews 2023 (Books)**
- **Designed a modular system architecture**

Plan for next phase

- **Expand evaluation with larger dataset**
- **Implement full system**
- **Conduct comprehensive analysis (Quantitative & Qualitative)**

Gantt Chart: Project Phases & Timeline



**THANK YOU FOR
LISTENING**

Appendix

System Requirements

Functional

- Accept text and/or image queries
- Retrieve semantically similar books
- Fuse multimodal signals
- Select appropriate NBA action type

Non-functional

- Handle missing modalities (image optional)
- Low-latency retrieval
- Modular & maintainable
- Research-friendly (easy to swap models)

Triplet Loss (distance based)

$$L_{triplet} = \max(0, d(a, p) - d(a, n) + m)$$

- Only enforce: “Positive should be closer than negative by a margin”
- Relative distances, not absolute similarity scores

Why Retrieval Alone Cannot Select the Right User Action

Retrieval Layer	NBA Policy Layer
Ranks items by relevance	Decides how to respond
Optimizes similarity	Optimizes user interaction
Always returns items	Can choose not to push a single item

Evidence from report:

- **NBA** is explicitly defined as **action selection**, not ranking (§1.1.2)
- **Policy engine** outputs **action types**, not item lists (§6.5)

NBA Policy Engine – Example Decision Trace

INPUT SIGNALS	Score
Fusion confidence	0.91
Score gap (top-1 vs top-2)	0.18
Modality balance	Text-dominant

POLICY INTERPRETATION

1. High confidence
2. Clear separation
3. Stable semantic match

SELECTED ACTION:

DIRECT RECOMMENDATION

**Full policy logic and calibration described in Report §6.5*

End